COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010/ Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	July 30 2026	**DATE SUBMITTED:**	July 30 2026
NAME:	Regala, Joaquin Luiz B.	**INSTRUCTOR:**	Engr. Jimlord Quejado

Hands-on Activity 4.1 [ACTIVITY NAME]

1. PROCEDURE OUTPUT

SCREENSHOTS

MAIN

```
#include <iostream>
#include "stackList.h"

int main(){

    std::cout<<"\n==== PUSHING 'c' 'p' 'e' ON TO THE STACK ====\n";

    push('c');
    push('p');
    push('e');

    Top<char>();

    std::cout<<"\n=== DISPLAY ALL ====\n";
    displayAll<char>();

    std::cout<<"\n==== POPPING ALL THE ELEMTENTS ====\n";

    while(!isEmpty<char>()){
        std::cout<<" Popped: "<< pop<char>() <<std::endl;
    }

    std::cout<<"\n==== TESTING THE UNDERFLOW ====\n";
    pop<char>();



    return 0;
}
```

stackList.h

```
#ifndef STACK_LIST_H
#define STACK_LIST_H

template <typename T>
class Node{
    public:
    T data;
    Node* next;
};
template <typename T>
Node<T> *head = nullptr;

template <typename T>
void push(T newData){

    Node<T>* newNode = new Node<T>;
    newNode->data = newData;

    newNode->next = head<T>;
    head<T> = newNode;
}
template <typename T>
T pop(){

    if (head<T> == nullptr){
        std::cout<<"Stack underflow"<<std::endl;
    }

    Node<T>* temp = head<T>;
    T tempVal = temp->data;
```

```
31      head<T> = head<T>->next;
32      delete temp;
33      return tempVal;
34
35  }
36  template<typename T>
37  void Top(){
38      if(head<T> == nullptr){
39          std::cout<<"Stack unerflow" <<std::endl;
40          return;
41      }
42      std::cout<<"Top of stack is "<<head<T>->data<<std::endl;
43  }
```

```
44  template<typename T>
45  bool isEmpty(){
46      return head <T> == nullptr;
47  }
48  template<typename T>
49  void displayAll(){
50      if(head<T> == nullptr){
51          std::cout<< "Stack underflow\n";
52          return;
53      }
54
55      std::cout<<"Stack from Top to Bottom: ";
56      Node<T>* current = head<T>;
57      while(current != nullptr){
58          std::cout<<current->data;
59          if(current->next != nullptr) std::cout<<" -> ";
60          current = current->next;
61      }
62      std::cout<<std::endl;
63  }
64  #endif
```

OUTPUT

```
==== PUSHING 'c' 'p' 'e' ON TO THE STACK ====
Top of stack is e

=== DISPLAY ALL ====
Stack from Top to Bottom: e -> p -> c

==== POPPING ALL THE ELEMTENTS ====
 Popped: e
 Popped: p
 Popped: c

==== TESTING THE UNDERFLOW ====
Stack underflow
```

COMPUTER ENGINEERING

QUEZON CITY

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Expression	Valid?(Y/N)	Output(Console Screenshot)	Analysis

Tool Analysis:

CONCLUSION:

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **NONE**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **NONE**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **NONE**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.